



UNIVERSITÀ DI PISA

Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

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Outline

1. Cuprates and Hubbard physics
2. The Hartree-Fock method
3. The normal phase
4. The antiferromagnetic phase
5. The superconducting phase
6. Conclusions and outlook

Cuprates and Hubbard physics

Nobel Prize in Physics 1987



Photo from the Nobel Foundation archive.

J. Georg Bednorz

Prize share: 1/2



Photo from the Nobel Foundation archive.

K. Alexander Müller

Prize share: 1/2

Possible High T_c Superconductivity in the Ba–La–Cu–O System

J.G. Bednorz and K.A. Müller

IBM Zürich Research Laboratory, Rüschlikon, Switzerland

Received April 17, 1986

What makes cuprates interesting?

- ◇ Strong “unconventional” insulators;
- ◇ Rich phase diagrams;
- ◇ High- T_c superconductivity;
- ◇ Unclear origin of Cooper pairing.

The timeline of high- T_c superconductivity

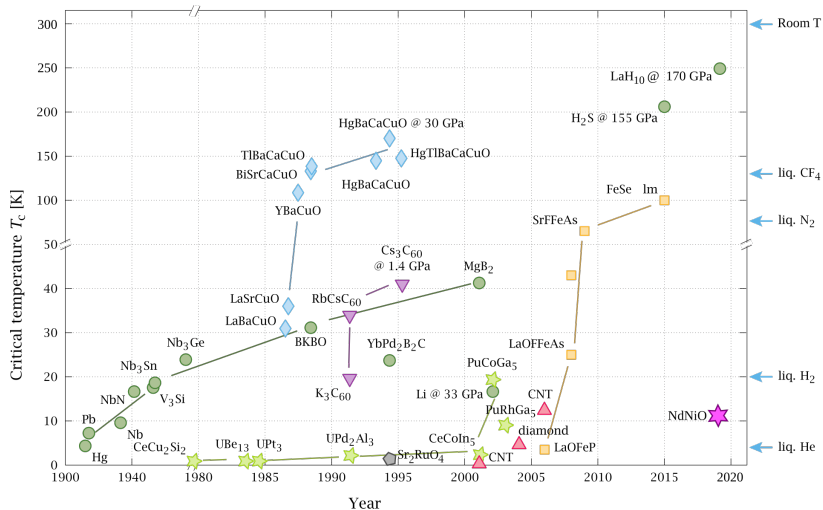
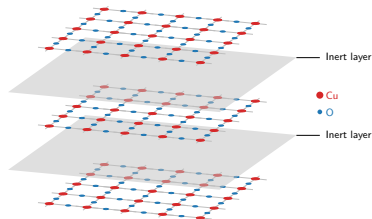


Figure 2: Timeline of the discovery of superconductivity in various compounds. The cuprates branch is represented as cyan diamonds. Image source: P. Ray and O. Gingras, distributed according to [the license CC BY-SA 4.0](https://creativecommons.org/licenses/by-sa/4.0/).

These materials are stackings of:

- ◇ 2D copper oxide active layers;
- ◇ intermediate layers of inert material.



Single-band Hubbard model on the square lattice:

$$\hat{H}_{\text{HM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{On-site repulsion}} \quad \text{with } t, U > 0.$$

The phase diagram of cuprates

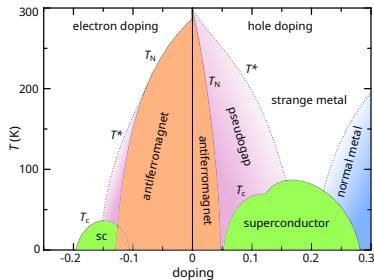
Within cuprates phase diagram:

- ◇ **Antiferromagnetism** at null and low doping;
- ◇ ***d*-wave superconductivity** as we inject or remove charge.

Both arise from **strong electronic correlations**.



Failure of band theory in describing insulation.



Schematic phase diagram of cuprates. Author: Holger Motzkau, [Wikimedia commons](#), CC BY-SA 3.0.

Mott localisation: a many-body insulator

What is the origin of the insulation mechanism?

Strong $e-e$ interactions and **big energetic cost** of having two electrons on the same site.

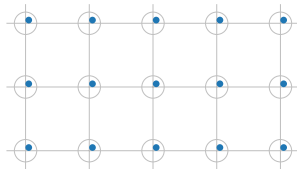
Strongly coupled Hubbard model:

$$U \gg t.$$



Mott insulation mechanism

Electrons are **localised** as in a sort of “traffic jam”.



Mott localisation: a many-body insulator

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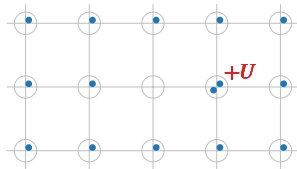
Strongly coupled Hubbard model:

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Mott insulation mechanism

Electrons are **localised** as in a sort of “traffic jam”.



How does Mott localisation relate to antiferromagnetism and superconductivity?

Antiferromagnetism

Mott localisation at half-filling.



Singlet pairing $\leftrightarrow$ AF ordering.



HM at strong coupling:

$$\hat{H}_{\text{HM}} \simeq \frac{4t^2}{U} \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \Delta E.$$

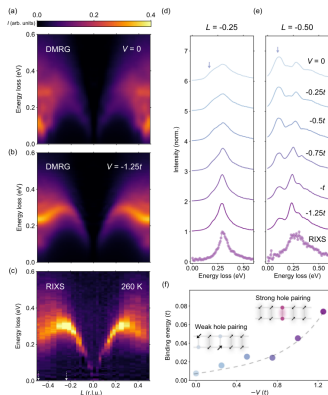
Superconductivity

Condensation of Cooper pairs with *d*-wave spatial structure.



Suppressed on-site pairing.

Recent developments: the extended Hubbard model



Hari Padma et al., 2025. *Physical Review X* 15 (2): 021049.

Incompatibilities with the spectrum of magnetic excitation in cuprates.



Proposal: extension of the model with a **non-local attraction**,

$$\hat{H}_{\text{EHM}} = \hat{H}_{\text{HM}} - V \underbrace{\sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\hat{H}_V}.$$

$\hat{H}_V$ can act as a **source of d -wave superconductivity**.

The model (EHM):

$$\hat{H}_{\text{EHM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{Local repulsion}} - \underbrace{V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\text{Non-local attraction}} \quad \text{with } t, U, V > 0.$$

The motivation: Compatibility with recent experimental findings.

The idea: we know that the EHM can host d -wave superconductivity. Can we obtain Mott physics?

The method: theoretical and numerical Hartree-Fock technique.

The Hartree-Fock method

The constrained minimisation framework

We want to approximate the true ground state $|\Psi_0\rangle$ of the EHM, with energy E_0 .

Variational principle:

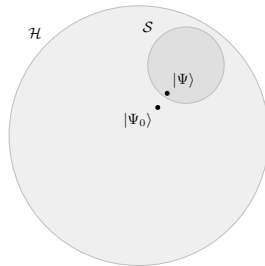
$$\forall |\Phi\rangle \in \mathcal{S} \subset \mathcal{H} : E[\Phi] \geq E_0.$$

$\Downarrow$

Constrained minimisation procedure

$$|\Psi\rangle : \left. \frac{\delta E[\Phi]}{\delta |\Phi\rangle} \right|_{|\Phi\rangle=|\Psi\rangle} = 0.$$

The state $|\Psi\rangle$ is the best approximation to $|\Psi_0\rangle$ inside of $\mathcal{S}$.



Sketch of the Hilbert space $\mathcal{H}$ and a subset $\mathcal{S}$.

The Hartree-Fock subspace of states

$\mathcal{S}$ is the set of all **gaussian states**,

$$|\Psi\rangle = \bigotimes_{\alpha} \hat{\gamma}_{\alpha}^{\dagger} |\Omega_{\gamma}\rangle, \quad \left\{ \hat{\gamma}_{\alpha}, \hat{\gamma}_{\beta}^{\dagger} \right\} = \delta_{\alpha\beta},$$

with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.



Each state can be identified **uniquely** by choosing

$$\Delta_{\alpha\beta}^{+-} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta} \rangle, \quad \Delta_{\alpha\beta}^{++} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle.$$

These are the **variational parameters** collected in the vector $\mathbf{v}$. We aim to find the optimal $\mathbf{v}$, such that

$$|\Psi[\mathbf{v}]\rangle \simeq |\Psi_0\rangle.$$

Phase transition: spontaneous breaking of some symmetries of the model.

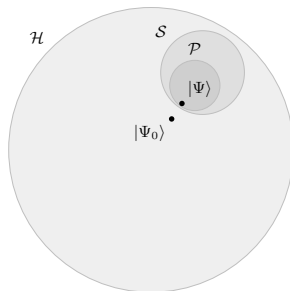
Phase X : shows symmetries $Y, Z \dots$



Further constraining of HF subspace to $\mathcal{P} \subset \mathcal{S}$, states with symmetries $Y, Z \dots$



Reduction of complexity on $\mathbf{v}$.



The model (EHM):

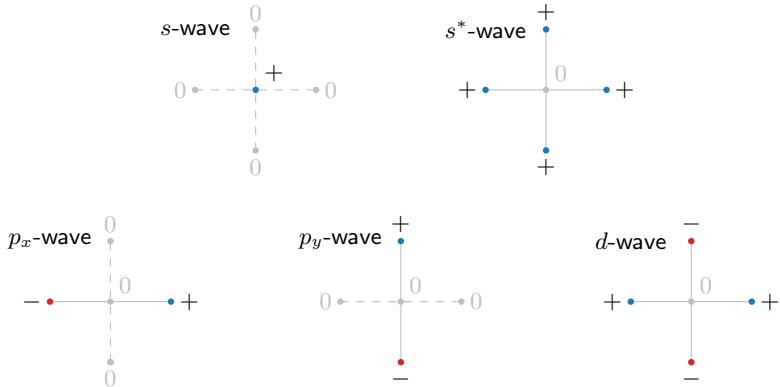
$$\hat{H}_{\text{EHM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{Local repulsion}} - \underbrace{V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\text{Non-local attraction}} \quad \text{with } t, U, V > 0,$$

and its symmetries:

- ◊ *Gauge invariance* (charge conservation), $U(1)$.
If broken: superconductivity.
- ◊ *Global spin rotational invariance*, $SU(2)$.
If broken: magnetism.
- ◊ *Translational invariance* in directions x and y , $T_1^{(x)} \otimes T_1^{(y)}$.
If broken: non-uniform charge and spin densities.
- ◊ *Rotational invariance* under $\pi/2$ rotations, C_4 .
If broken: nematicity.

The spatial symmetry channels

It is useful to introduce five spatial symmetry channels:



The normal phase

The self-consistency problem for the normal phase

Normal phase: no broken symmetries. Just one variational parameter,

$$\mathbf{v} = \left[u^{(s^*)} \right] ,$$

where $u^{(s^*)}$ is the s^* -wave component of $u_{\mathbf{k}\sigma} \equiv \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle$.

$\Downarrow$

Kinetics renormalisation

Hopping amplitude is **isotropically renormalised**,

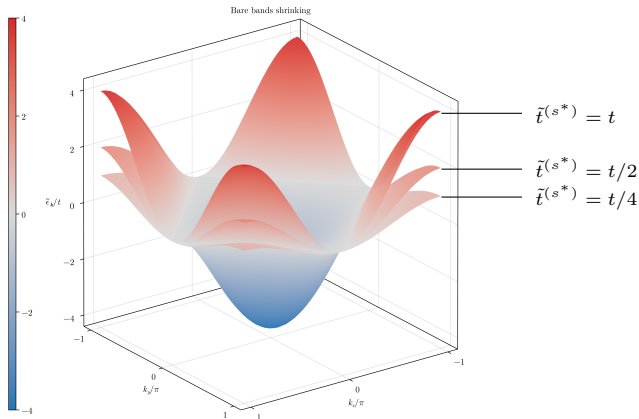
$$t \rightarrow \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)} V}{2} .$$

$\Downarrow$

Renormalisation of electronic bands, $\epsilon_{\mathbf{k}\sigma} \rightarrow \tilde{\epsilon}_{\mathbf{k}\sigma}$.

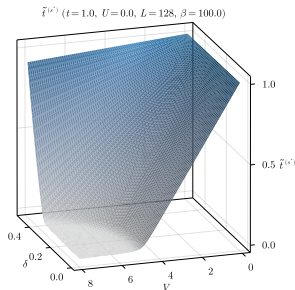
Theoretical constraint:

$$0 < \tilde{t}^{(s^*)} < t \quad \Rightarrow \quad \text{Flattening of bands} = \text{Mott localisation.}$$

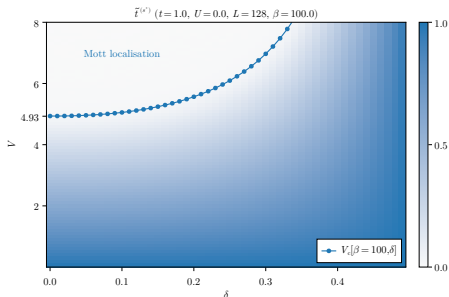


Complete flattening of bare bands

In (δ, V) plane, we have determined theoretically the presence of a Mott critical boundary $V_c[\beta, \delta]$,



(a) Numerical results for $\tilde{\epsilon}(s^*)$



(b) Critical Mott boundary.

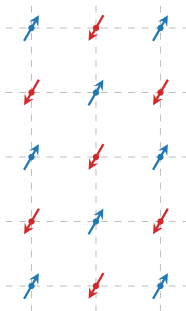
Within the Mott plateau, electrons are completely localised.

The **normal phase**, summarised:

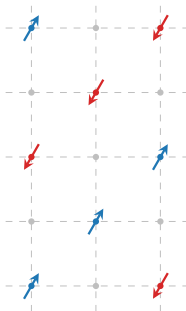
- ◇ Preserves all symmetries of the EHM;
- ◇ The HF approximation is **independent of U** ;
- ◇ Presence of V -induced Mott localisation and **bands flattening**;
- ◇ We identify a Mott transition boundary as we vary the doping level.

The antiferromagnetic phase

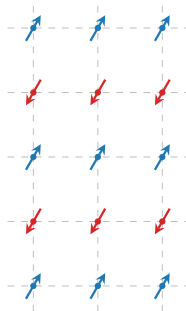
Among all possible realisations of AF ordering,



Alternating



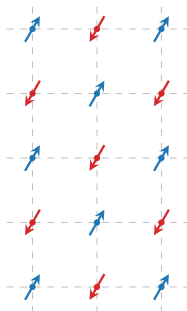
Doubled Periodicity



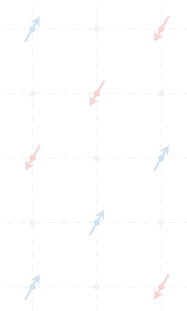
Stripes

we restrict to “commensurate antiferromagnetism”.

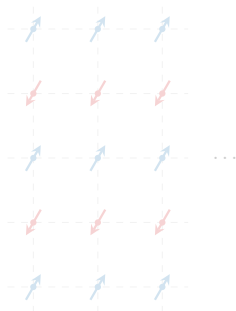
Among all possible realisations of AF ordering,



Alternating



Doubled Periodicity



Stripes

we restrict to “commensurate antiferromagnetism”.

AF phase: lowers $SU(2)$ and $T_1^{(x)} \otimes T_1^{(y)}$.

In the HM ($V = 0$), the only HFP is the magnetisation m .



U -induced magnetisation, disrupted by δ .

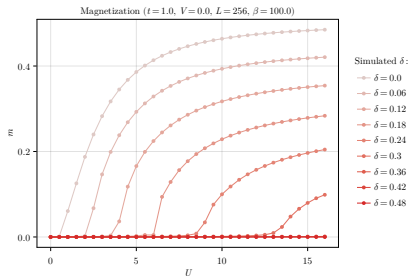


Figure 5: Numerical results for m .

Half-filling restriction

At any finite filling ($\delta \neq 0$) the commensurate AF phase is **unstable**. We limit to $\delta = 0$ (half-filling).

Mott localisation in the AF phase

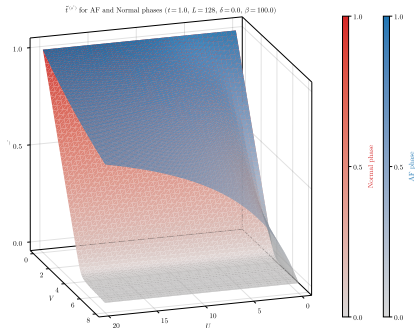


Figure 6: Comparative plot of the converged values for $\tilde{t}^{(s^*)}$ for the normal phase and the AF phase.

Setting $V > 0$, we get the new HFP $u^{(s^*)}$ as for the normal phase,

$$t \rightarrow \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2}.$$

Theoretical constraint:

$$0 < \tilde{t}^{(s^*)} < t$$



Mott localisation in the AF phase

The local repulsion U **opposes bands flattening** $\implies$ counter-intuitive!

Magnetisation is **enhanced** by Mott localisation:

$$0 < \tilde{t}^{(s^*)} < t \quad \Longrightarrow \quad U/\tilde{t}^{(s^*)} > U/t.$$

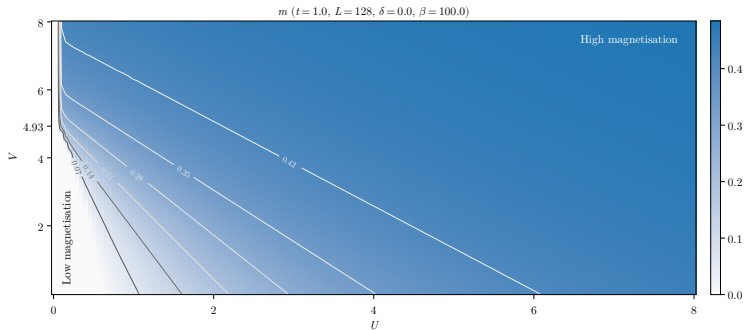


Figure 7: Numerical results for the HFP m . The gray curves indicate isomagnetic lines.

The **AF phase**, summarised:

- ◇ Breaks spin rotation and translational invariance;
- ◇ The HF approximation is only stable at half-filling;
- ◇ Presence of V -induced Mott localisation and **bands flattening**, enhanced by V and opposed by U ;
- ◇ Two ways of magnetising the lattice: increasing U ("conventional") or using Mott effects ("unconventional");
- ◇ (Not shown in slides) always energetically favoured compared to the normal phase.

The superconducting phase

SC phase: breaking of charge conservation $U(1)$, insurgence of a SC gap.



Anomalous densities can be non-zero,

$$\Delta_{\alpha\beta}^{++} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle \neq 0.$$

HFPs of the SC phase

Harmonics of the conventional density,

$$u_{\mathbf{k}} \equiv \langle \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma} \rangle \rightarrow u^{(\gamma)}.$$

Harmonics of the anomalous density,

$$w_{\mathbf{k}} \equiv \langle \hat{c}_{\mathbf{k}\uparrow}^{\dagger} \hat{c}_{-\mathbf{k}\downarrow}^{\dagger} \rangle \rightarrow w^{(\gamma)}.$$

Gap symmetries in the singlet channel

Singlet gap $\tilde{\Delta}_s$ harmonics ($t = 1.0$, $U = 4.0$, $L = 128$, $\beta = 100.0$)

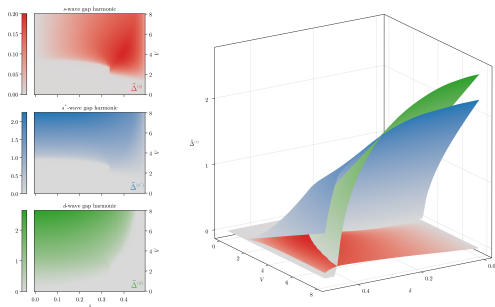


Figure 8: Singlet gap harmonics $\tilde{\Delta}^{(\gamma)}$, with $\gamma = s, s^*, d$ at $U/t = 4$.

Gap components in
singlet channel:

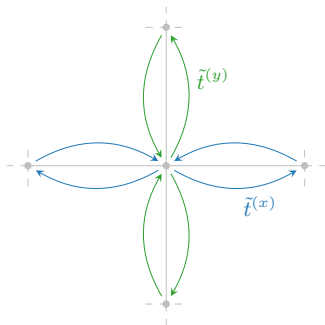
$$\begin{aligned}\tilde{\Delta}^{(s)} &= -Uw^{(s)}, \\ \tilde{\Delta}^{(s^*)} &= Vw^{(s^*)}, \\ \tilde{\Delta}^{(d)} &= Vw^{(d)}.\end{aligned}$$

Internal constraint among singlet harmonics

$$V \sum_{\gamma \neq s} |w^{(\gamma)}|^2 \geq U |w^{(s)}|^2.$$

In the SC phase, the renormalisation of t gives two components,

$$t \rightarrow \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2} \quad \text{and} \quad \tilde{t}^{(d)} \equiv -\frac{u^{(d)}V}{2}.$$



Anisotropic hopping

Direction-resolved effective hopping:

$$\tilde{t}^{(x)} \equiv \frac{\tilde{t}^{(s^*)} + \tilde{t}^{(d)}}{2},$$

$$\tilde{t}^{(y)} \equiv \frac{\tilde{t}^{(s^*)} - \tilde{t}^{(d)}}{2}.$$

$\Downarrow$

$\tilde{t}^{(d)} \neq 0 \implies$ **nematic SSB.**

Symmetry mixing and insurgence of nematicity

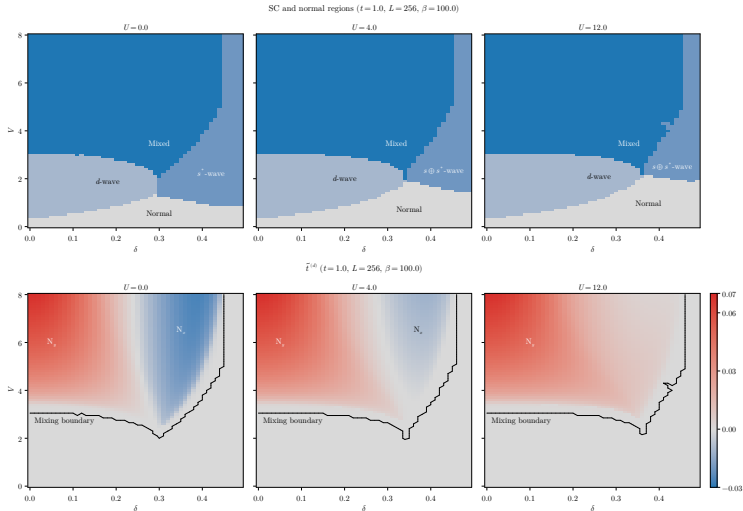
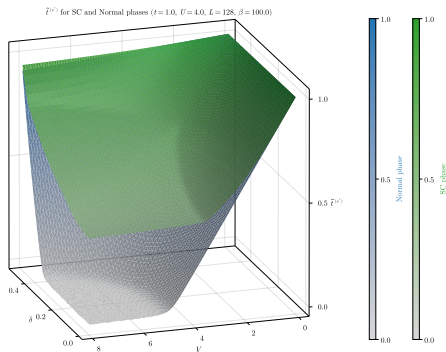


Figure 9: Comparison of phase diagrams for the singlet SC phase and convergence values for the nematic order parameter.



Isotropic hopping
renormalisation:

$$\tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2}.$$

Formally identical to
normal and AF phases.

Figure 10: Comparative plot of the converged values for $\tilde{t}^{(s^*)}$ for the normal phase and the SC phase.

Isotropic flattened component $\gg$ Anisotropic correction.

The **SC phase**, summarised:

- ◇ Breaks charge conservation and rotational symmetry;
- ◇ Symmetry mixing in singlet channel;
- ◇ Mott localisation and nematicity have the same origin;
- ◇ (Not shown in slides) always energetically favoured compared to the normal phase;
- ◇ (Not shown in slides) favoured at $V \gg U$ compared to the AF phase.

Conclusions and outlook

The idea: we know that the EHM can host d -wave superconductivity. Can we obtain Mott physics?

At HF level, the non-local attraction:

- ◇ Acts as a source of Mott localisation;
- ◇ Enhances magnetisation at half filling;
- ◇ Can produce mixed-symmetry superconductivity;
- ◇ Can produce nematic superconductivity.

From here, we can follow various paths:

1. *Same model, other phases.*

Triplet superconductivity, breaking of translational invariance, nematic antiferromagnetism. . .

2. *Expand the model.*

Second nearest neighbours hopping, more refined potentials. . .

3. *More advanced numerical techniques.*

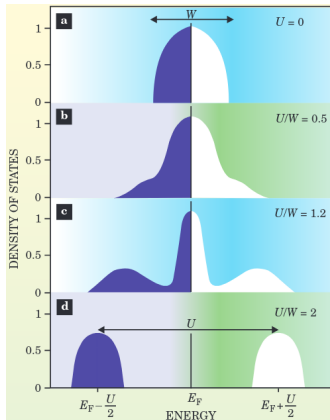
DMRG for chains and ladders; hybrid DMFT (local sector) and HF (non-local sector) to retrieve Mott physics induced by U . . .

or any linear combination of the three!

That's all, thank you!

Backup slides

Bands flattening: a signature of Mott localisation



Gabriel Kotliar and Dieter Vollhardt, 2004.
Physics Today 57 (March): 53–59.

As we increase the ratio U/t , the many-body DoS undergoes a transition:

- (a) Free model, conventional DoS;
- (b) Weak coupling, large coherent peak;
- (c) Intermediate coupling, **narrow coherent peak**;
- (d) Strong coupling, lower and upper Hubbard bands.



Localisation = flattening of bare bands.

Hartree-Fock method: $\mathcal{S}$ is the set of all **gaussian states**, i.e. states that are non-interacting for some fermionic operators

$$|\Psi\rangle = \bigotimes_{\alpha} \hat{\gamma}_{\alpha}^{\dagger} |\Omega_{\gamma}\rangle, \quad \{\hat{\gamma}_{\alpha}, \hat{\gamma}_{\beta}^{\dagger}\} = \delta_{\alpha\beta},$$

with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.



Wick's theorem: averages of quartic operators decompose nicely,

$$\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \hat{c}_{\delta} \rangle = \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\delta} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\gamma} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\delta} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle \langle \hat{c}_{\gamma} \hat{c}_{\delta} \rangle}_{\text{Cooper}}.$$

The EHM hamiltonian is a quartic operator!

Each $|\Psi\rangle \in \mathcal{S}$ is uniquely defined by a particular choice of the **variational parameters** (HFPs)

$$\Delta_{\alpha\beta}^{+-} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta} \rangle, \quad \Delta_{\alpha\beta}^{++} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle.$$

Let $\mathbf{v}$ be the vector that collects all HFPs. We aim to find

$$|\Psi[\mathbf{v}]\rangle \simeq |\Psi_0\rangle,$$

with $\mathbf{v}$ such that:

Variational picture

Energy is minimised,

$$\left. \frac{\delta E[\Phi]}{\delta |\Phi\rangle} \right|_{|\Phi\rangle=|\Psi[\mathbf{v}]\rangle} = 0.$$

Self-consistency picture

A self-consistency problem is satisfied,

$$A(\mathbf{v}) = \mathbf{v},$$

being A a non-linear map **uniquely determined**.

The iterative algorithm (iHF)

The HF solution is the **fixed point** of a non-linear map A ,

$$A(\mathbf{v}) = \mathbf{v}.$$



Strategy: “follow” the map up to the fixed point,

$$\mathbf{v}_{i+1} \equiv gA(\mathbf{v}_i) + (1 - g)\mathbf{v}_i.$$

with $i > 0$, $g \in [0, 1]$.

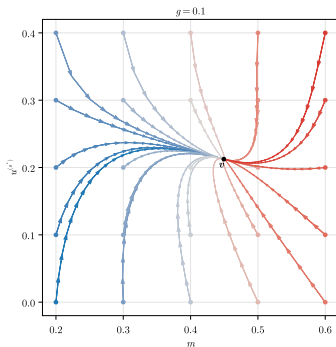


Convergence criterion:

$$\forall j \quad : \quad [\mathbf{v}_{i+1} - \mathbf{v}_i]_j < \Delta v_j,$$

with $\Delta \mathbf{v}$ the vector of tolerances.

Example path (AF) ($t = 1.0$, $U = 3.0$, $V = 6.0$, $L = 128$, $\beta = 100.0$, $\delta = 0.0$)



Example path for the AF phase.